

Support Vector Machine (SVM) - Task 9 Explanation

Introduction: Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that separates data points of different classes.

Hyperplane: A hyperplane is a decision boundary that separates data points. In 2D, it is a line; in higher dimensions, it becomes a plane or hyperplane.

Margin: Margin is the distance between the hyperplane and the nearest data points (support vectors). SVM aims to maximize this margin.

Kernel Trick: The kernel trick allows SVM to perform classification in higher-dimensional space without explicitly transforming the data. Common kernels include Linear and RBF (Radial Basis Function).

Linear Kernel: Used when data is linearly separable. It creates a straight decision boundary.

RBF Kernel: Used for non-linear data. It maps data into higher dimensions to find a separating hyperplane.

Model Evaluation: Confusion matrix is used to evaluate model performance by comparing predicted and actual values.

Conclusion: SVM is powerful for high-dimensional data and effective in cases where number of features is greater than samples.